

**Diesel Engines****V956TB33****V1163TB33****Maintenance Schedule****M050253/11E**

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Low operating and maintenance costs as well as operational reliability and availability depend on maintenance and servicing being carried out in compliance with our specifications and instructions.

The overall system, of which the engine is an integral part, must be maintained in such a way as to ensure trouble-free engine operation at all times. For this purpose always:

- ensure that sufficient fuel is available,
- ensure that the combustion air is dry and clean,
- use only dry, clean compressed air,
- use only clean, filtered raw water.

Moreover, it is essential that:

- maintenance tasks be completed by trained personnel,
- suitable tools be used,
- genuine spare parts and fluids as per the current Fluids and Lubricants Specification be used.

Our Product Support Service will always be available should assistance be required.

Preventive maintenance instructions.

- Special care should be taken to keep the machinery plant in a clean and serviceable condition at all times to facilitate early detection of possible leaks and prevent subsequent damage.
- Protect rubber and synthetic components from oil and fuel, never treat with organic detergents. Wipe with a dry cloth only.

Maintenance Concept

The maintenance concept for mtu products is a preventive maintenance concept and features the maintenance echelons W1 to W6 as outlined below.

Preventive maintenance facilitates advance planning and ensures a high degree of equipment availability.

Our Operational Data Recorder (ODR) allows maintenance for tasks marked "•" to be carried out not only as scheduled but according to component condition (on-condition maintenance).

On-condition maintenance reduces costs.

Main Features of Maintenance Tasks for mtu engines.

Maintenance Echelon W1: Operational Checks.

Maintenance Echelons W2, W3 and W4: Periodic maintenance tasks to be performed during out-of-service periods without the need for engine disassembly.

Maintenance Echelon W5: This echelon requires partial engine disassembly.

Maintenance Echelon W6: Major overhaul. This echelon requires complete engine disassembly.

The time intervals according to which the W2 to W6 Maintenance Echelons and the relevant checks and tasks involved are to be completed are based on operational experience. They have been determined so as to ensure correct engine operation until the next scheduled maintenance echelon.

Specific operational conditions may require modification of the Maintenance Schedule.



Maintenance Schedule

Group	D
Subgroup	000
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NOTE

The codes on the following pages refer to the groups and subgroups of the Task Descriptions in Section G.

This Maintenance Schedule may refer to parts which are not installed on your engine, these may be disregarded.





Application Group

3B: Electric power generation, continuous duty (75% average utilization based on operating time)

3C: Electric power generation, continuous duty (annual limit)

Maintenance Frequency Chart

	3B	3C
Maintenance Echelon W1: Operational checks	Daily	Daily
W2: Operating hours	250	250
Limit, months	6	6
W3: Operating hours	1 000	1 000
Limit, year	1	1
W4: Operating hours	2 000	1 500
Limit, year(s)	2	2
W5: Operating hours	6 000	3 000
Limit, year(s)	6	4
W6: Operating hours	18 000	9 000
Limit, year(s)	18	18

Maintenance Task Schedule

Code No.	One-time operations after the first 50 operating hours - With a new engine, after W5 maintenance echelon, or after W6 major overhaul	
G00.049	Attachments	Check tightness of securing screws and nuts
G06.101	Valve gear	Check valve clearances, adjust if necessary
G12.311	Fuel pre-filter	Replace filter elements
G12.311	Fuel pre-filter	Clean
G12.321	Fuel duplex filter	Replace filter elements
G14.019	Engine coolant system	Clean strainer before coolant thermostat. Check mesh for damage
G16.002	Engine oil	Take sample and analyse

Code No.	Maintenance Echelon W1: -- Operational checks	
G10.051	Exhaust system	Check exhaust gas colour Drain condensate (if drain valve is provided)
G10.181	Intake air line	Check drain line for water discharge and obstructions
G10.211	Air filter	Check contamination indicator
G12.311	Fuel pre-filter	Check differential pressure
G14.011	• Engine coolant	Check level
G14.051	Charge air coolant	Check level
G16.002	• Engine oil	Check level
G16.111	Engine oil filter	Operate filter handle several times
G84.001	Monitoring system	Carry out lamp test
G84.002	Engine operation	Check running noises Check engine and external pipework for leaks Check speeds, pressures and temperatures - if gauges are provided
G86.051	Compressed air system	Check operating pressure Drain condensate